9 $T = \sqrt{\dfrac{w}{d^3}}$

$w = 5.6 \times 10^{-5}$
$d = 1.4 \times 10^{-4}$

(a) Work out the value of T.
Give your answer in standard form correct to 3 significant figures.

$d^3 = (1.4 \times 10^{-4})^3 = 2.744 \times 10^{-12}$

$w \div d^3 = \dfrac{5.6 \times 10^{-5}}{2.744 \times 10^{-12}} = 2.040816327... \times 10^7$

1 mark
Calculate the quantity inside the square root

$T = \sqrt{(2.040816327... \times 10^7)} = 4517.5395...$

$T = 4.52 \times 10^3$

1 mark
Standard form, rounded to 3 significant figures

(2)

w is increased by 10%
d is increased by 5%

Lottie says,

"The value of T will increase because both w and d are increased."

(b) Lottie is wrong.
Explain why.

A 10% increase multiplies w by 1.10.

A 5% increase multiplies d by 1.05, so d^3 by 1.05^3.

Multiplier for the value inside the root:

$\dfrac{1.10}{1.05^3} = 1.10 \div 1.157625 < 1$

1 mark
Use the percentage changes and the cube of d

Multiplier for T = $\sqrt{(1.10 \div 1.05^3)} \approx 0.974793$.

This is less than 1, so T decreases.

The denominator grows proportionally more than the numerator.

(2)

1 mark
Support the decrease with a multiplier below 1

(Total for Question 9 is 4 marks)